

Akshat Kankani

+91 8420708412 | kankaniakshat185@gmail.com | [linkedin.com/in/akshat185](https://www.linkedin.com/in/akshat185) | github.com/kankaniakshat185 | akshatkankani.vercel.app

SUMMARY

Computer Science undergraduate (Artificial Intelligence specialization) focused on systems programming, backend infrastructure, and performance-critical software. Built systems in C++ and Python spanning storage engines, networking, concurrency, and performance optimization.

EDUCATION

Manipal Institute of Technology

B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.11

Bengaluru, KA

July 2023 – May 2027

PROJECTS

Strata | C++20, POSIX APIs (file I/O, fsync, signals), Multithreading, Make, GitHub Actions

GitHub

- Built a tiered-resolution time-series database in C++20 (2,400 LOC, zero third-party dependencies), implementing a write-ahead log, MemTable, and LSM-style compaction pipeline.
- Engineered a bit-level Gorilla compression codec (Facebook's algorithm: delta-of-delta timestamps, XOR'd values) from the published paper, reducing storage requirements by **4.3×**.
- Designed an inverted index for label-based queries, keeping p50 lookup latency sub-**500ns** as the index scaled from 1K to 1M unique series.
- Proved crash safety via deterministic fault injection, killing the process at 5 exact code points mid-write/mid-compaction with zero data loss.

Low-Latency Matching-engine | C++20, CMake, GoogleTest, GitHub Actions

GitHub

- Built a limit order book matching engine in C++20 implementing price-time priority, partial fills, and O(1) cancellation for limit and market orders.
- Engineered four interchangeable order-book implementations behind one templated engine, boosting throughput 294% to 12.33M actions/sec.
- Validated correctness via differential testing and a 20,000-op fuzz test checking book invariants, reaching 71 tests and 99.2% line coverage.
- Automated a 5-job GitHub Actions CI pipeline with sanitized builds, static analysis, and coverage reporting, gating every push and pull request.

Custom-Http-server | Python, HTTP Protocols, Sockets, Docker, Concurrency & Multithreading

GitHub

- Built a modular, multithreaded HTTP/1.1 framework in Python with middleware chaining, dynamic routing, and gzip compression, hitting 1,198 req/sec, 17% faster than Flask.
- Diagnosed and fixed a server-shutdown deadlock and a socket selector race condition via single-writer locking, cutting shutdown under load from a hang to 0.5s.
- Hardened against path-traversal and DoS attacks via canonical path validation, 16KB/10MB request-size limits, and idle-connection timeouts.
- Engineered a 33-test pytest suite spanning routing, parsing, and security, gated by GitHub Actions CI, and containerized with Docker for deployment.

TECHNICAL SKILLS

Languages: C++, Python, SQL

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning

Databases: MongoDB, PostgreSQL

Developer Tools: Git, GitHub, Linux, Postman, Pytest, GitHub Actions

Backend & Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, Redis, Celery, Sockets

ML & AI: NumPy, Pandas, Matplotlib, scikit-learn

ACHIEVEMENTS & CERTIFICATIONS

- Solved **250+** Data Structures & Algorithms problems in **C++** across LeetCode and other coding platforms.
- Achieved a peak **LeetCode Contest Rating of 1862 (Knight)** through consistent participation in weekly and biweekly coding contests.
- Completed the **AWS Cloud Solutions Architect Professional Certificate** on Coursera.